🚀 ESSENTIOS MVP ROADMAP (7-14 DAYS)

On-Chain LLM Agent Implementation Plan

By: Manus AI (Based on Gorilla's Blueprint) **Goal:** Deploy the Minimum Viable Product (MVP) of the Essentios Agent, the Blockchain Brain Assistant.

Introduction: The 14-Day Sprint

This plan is a 14-day sprint designed to validate the three pillars of Essentios: **AI Understanding**, **Secure Execution**, and **Tamper-Proof Memory (Witness Journal)**.

Each step is clearly defined with a success criterion to ensure fast and measurable progress.

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1. Phase 1: Foundations (Days 1 to 3)

This phase establishes the working environment and the on-chain memory contract.

Day(s)	Main Objective	Key Task(s)	Success Criterion
Days 1-2	**Project Setup & Environment**	Initialize Rust (CosmWasm) and Python (Agent CLI) workspaces. Set up the local development environment (junod or wasmd).	Repository skeletons created. Local chain is running.
Day 3	**Witness Journal Contract (Skeleton)**	Implement the `witness_journal` CosmWasm contract with `AppendLog` and `GetLogs` functions. Define the administrator (the agent).	Contract compiles. Unit tests for adding and querying logs pass.

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2. Phase 2: Brain and Language (Days 4 to 6)

This phase gives the agent its ability to understand human language and translate it into transaction drafts.

Day(s)	Main Objective	Key Task(s)	Success Criterion
Day 4	**Agent CLI – Transaction Drafting**	Code the CLI tool (Python) to query chain state (balance, proposals). Implement a simple command parser (e.g., "Stake 5 TOKEN").	The Agent CLI outputs a plausible transaction JSON for at least one simple command.
Day 5	**LLM Integration (Local)**	Integrate a local LLM (e.g., LLaMA2 7B via Ollama) for intent parsing. Use a JSON schema prompt to structure the output.	The agent converts at least 3 complex intents (stake, transfer, proposal) into correct transaction JSON.
Day 6	**Cosmos Tx Execution Flow**	Implement signing and broadcasting. Use GRPC to **simulate** the transaction first (dry-run) before asking for confirmation.	The agent can send a basic transaction (e.g., token transfer) after user confirmation, and the transaction is included in the block.

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3. Phase 3: Transparency and Testing (Days 7 to 9)

This phase connects the brain to the tamper-proof memory and tests the governance scenario.

Day(s)	Main Objective	Key Task(s)	Success Criterion
Day 7	**On-Chain Logging & Transparency**	Deploy the `witness_journal` contract. After each confirmed action, the agent calls `AppendLog` on the contract.	The on-chain log length increases by 1 after every confirmed agent action.
Day 8	**DAO Proposal Demo Script**	Set up a governance test scenario (DAO). The agent must create a DAO proposal from a natural language command.	End-to-end demonstration: NL → Draft → Confirmation → DAO Proposal Submission.
Day 9	**Enhance Understanding & Context**	Implement chain state queries (e.g., "What's my reward balance?"). Use the LLM to format a user-friendly answer.	The agent correctly answers questions about the chain state in natural language.

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4. Phase 4: Refinement and Security (Days 10 to 14)

This phase focuses on improving security, memory, and user experience.

Day(s)	Main Objective	Key Task(s)	Success Criterion
Days 10-11	**Off-Chain & Contextual Memory**	Integrate off-chain storage (IPFS/Arweave) for long conversations. Use a lightweight Vector DB for semantic indexing.	The agent can recall the context of a previous conversation and summarize it.
Day 12	**Advanced Safety Filters**	Implement rule-based safety filters (e.g., spending limits, address whitelists) to intercept dangerous LLM outputs.	The filter blocks an unauthorized transaction attempt (e.g., transfer to a non-whitelisted address).
Days 13-14	**Documentation & Audit**	Finalize technical documentation and user guide. Conduct an internal audit of the `witness_journal` contract code.	Complete documentation. The `witness_journal` contract is audited and deemed safe.

Conclusion

This 14-day plan is the fastest path to bringing Essentios to life. It ensures that the principles of **Security**, **Transparency**, and **Human Control** are validated from the MVP version onwards.